

Savanna Sentinel

Brand Style Guide

Colour tokens, type scale, components, accessibility requirements. The visual system behind Savanna Sentinel. Each decision exists to answer one question: does this help a ranger read the data faster?



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The Golden Thread

This guide is a chain. Brand values shape voice. Voice shapes visual choices. Those choices become reusable patterns.

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Values	Voice	Visual	Patterns	Code
Relevant data over noise. A ranger needs to know what matters, not everything at once.	Calm and specific. A sighting and a poaching alert sound like the same person wrote them.	Dominant colour: Deep Navy. Typefaces: Barlow Condensed for display, Barlow for body. Spacing follows a 4pt grid. Status colours appear only when something is important.	Buttons, forms, alerts, navigation. Every component shows its state.	<code>--brand-primary</code> <code>--sp-4 / --dur-fast</code>

Brand Foundation

From the colour of a danger alert to the wording of an explanation, they all come from the same decisions. Everything else in this guide follows from what's here.

Savanna Sentinel gives wildlife rangers access to poaching data. This includes incident reports, heatmaps, patrol coverage, community tip-offs and more. All of it readable on a phone in the field.

Values

The platform shows important data clearly and keeps everything else out of the way. A ranger in the field can't afford ambiguity. The app fails if important data doesn't hold contrast in direct sunlight.

Voice

The platform sounds like someone who's giving a simple report. Specific, not corporate. It doesn't overstate alerts or sweeten success. The same voice tells a patrol summary and a poaching incident without switching registers.

Tone

The urgency changes, the voice doesn't. A patrol summary reads like a summary. An alert gets to the point.

Design Tokens

Every colour, spacing value, and duration in the app has a named CSS custom

property. Most components reference these tokens directly. A few precise pixel dimensions (such as decorative shadow overlays) intentionally use raw values where no exact token exists.

Colour Tokens

	TOKEN	VALUE	USAGE
☐	<code>—color-surface-bg</code>	#F2F2F2	Page canvas
☐	<code>—color-surface-raised</code>	#FFFFFF	Cards, panels, modals
☐	<code>—color-surface-deep</code>	#002354	Dark decorative backgrounds. Also the hover/pressed state for text that defaults to Brand Primary.
☐	<code>—color-border</code>	#D8DAD6	Dividers, input outlines, separators
☐	<code>—color-text-primary</code>	#313131	Body copy, 11.6:1 on surface-bg
☐	<code>—color-text-inverse</code>	#FFFFFF	All text on dark surfaces, 15.3:1 on surface-deep
☐	<code>—color-text-secondary</code>	#4F7392	Timestamps, metadata, secondary labels. On surface-raised at 14px+; on surface-bg at 24px+ only. Do not use for small text on surface-bg, except on disabled/inactive controls, which are exempt from contrast requirements.
☐	<code>—brand-primary</code>	#003A6B	Navigation sidebar, hero, primary buttons, headings on light, 10.3:1 on surface-bg
☐	<code>—brand-mid</code>	#174585	Interactive elements, links on light, 8.4:1 on surface-bg
☐	<code>—brand-steel</code>	#6897B8	Decorative on dark, large text on dark only

	TOKEN	VALUE	USAGE
☐	—brand-muted	#4F7392	Secondary labels at 24px+ on light, large text only
☐	—brand-light	#8EADC4	Decorative on dark only (insufficient contrast for UI text)
☐	—brand-teal	#056595	Decorative overlays, secondary icon backgrounds
☐	—status-safe	#06B050	Safe zones, sync success, patrol clear
☐	—status-caution	#FAE203	Low risk monitoring, non blocking warnings
☐	—status-alert	#FF9300	Elevated risk, requires attention
☐	—status-critical	#B30000	Immediate action required
☐	—spot-navy	#103364	Chart contrast, instructional diagrams
☐	—spot-cobalt	#2042B2	High visibility diagram elements, chart series
☐	—spot-sky	#0580BB	Chart series, diagram contrast alongside primary blues
☐	—spot-blue	#0070BF	Data differentiation in diagrams and charts
☐	—spot-aqua	#009193	Decorative overlays, chart series, diagram fills
☐	—status-safe-text	#0A7D3A	Safe state text on light surfaces
☐	—status-caution-text	#7A4800	Caution and alert state text on light surfaces (both yellow and orange fail WCAG as text. Use this token for both)
☐	—status-critical-text	#9B0000	Critical state text and form error messages on light surfaces

Spacing Tokens (4pt grid)

TOKEN	VALUE	EXAMPLE USAGE
–sp-half	2px	Micro-spacing. Badge vertical padding, small chip insets
–sp-1	4px	Small gaps. Icon to label, badge icon-to-label gap
–sp-2	8px	Tight spacing. Icon tile gap, badge row gap, map legend row gap, filter chip gap
–sp-3	12px	Input internal padding, route card padding, row gap
–sp-4	16px	Popup outer padding, form field gap
–sp-5	20px	Card padding, component internal padding
–sp-6	24px	Section inner padding, nav item spacing
–sp-8	32px	Two column grid gap, section outer margin
–sp-10	40px	Token group margin, proportion legend margin
–touch-target	44px	Interactive element touch size.
–sp-12	48px	Site nav height, mobile hero top padding
–topbar-height	56px	Top bar / app header height
–sp-16	64px	Section vertical padding, hero bottom
–sp-20	80px	Hero top padding, large section breathing room
–sp-24	96px	Maximum vertical rhythm

Motion Tokens

TOKEN	VALUE	EXAMPLE USAGE
<code>—dur-instant</code>	80ms	Data table row hover background transition
<code>—dur-fast</code>	150ms	Hover states, icon swaps, tooltips
<code>—dur-normal</code>	250ms	Modal and panel exit animations
<code>—dur-deliberate</code>	400ms	Page reveals, modal entrance, map overlays
<code>—dur-slow</code>	600ms	Loading spinner rotation
<code>—ease-standard</code>	cubic-bezier(0.2, 0, 0, 1)	Most UI transitions, balanced in/out
<code>—ease-enter</code>	cubic-bezier(0, 0, 0.2, 1)	Elements entering the screen (decelerate)
<code>—ease-exit</code>	cubic-bezier(0.4, 0, 1, 1)	Elements leaving the screen (accelerate)

Shape, Shadow, Z-Index Tokens

TOKEN	VALUE	EXAMPLE USAGE
<code>—radius-xs</code>	2px	Checkboxes, progress/confidence bars, focus outline corners
<code>—radius-sm</code>	4px	Close buttons, focus ring corners, skeleton loaders, pagination controls
<code>—radius-md</code>	8px	Buttons, inputs, small cards
<code>—radius-lg</code>	12px	Cards, panels, modals
<code>—shadow-sm</code>	0 1px 3px rgba(0,0,0,.10)	Cards at rest
<code>—shadow-md</code>	0 4px 12px rgba(0,0,0,.12)	Dropdowns, hovered cards
<code>—z-sticky</code>	200	Sticky headers, table columns

TOKEN	VALUE	EXAMPLE USAGE
—z-dropdown	300	Dropdown menus, popovers, select panels
—z-modal	400	Modals, drawers
—z-toast	500	Toast stack
—z-tooltip	600	Tooltips, popovers

Letter-spacing Tokens

TOKEN	VALUE	EXAMPLE USAGE
—ls-tight	-0.02em	Display headings, large stat values
—ls-wide	0.05em	UI element labels, form section labels
—ls-wider	0.5px	Overlines, caption text with uppercase
—ls-widest	1px	Column headers, focus tile labels
—ls-stat	1.5px	Stat card labels

Line-height Tokens

TOKEN	VALUE	EXAMPLE USAGE
—lh-none	1	Large numerics, stat card values
—lh-tight	0.95	Display headings only

TOKEN	VALUE	EXAMPLE USAGE
—lh-snug	1.25	Heading 4, subheadings
—lh-normal	1.5	UI labels, captions
—lh-relaxed	1.65	Body text (default), body large, toast body
—lh-loose	1.7	Section leads, reading paragraphs

Font-weight Tokens

TOKEN	VALUE	EXAMPLE USAGE
—fw-normal	400	Body text, body large
—fw-medium	500	Buttons, form labels, nav items, UI labels
—fw-semibold	600	Overlines, column headers, secondary labels, badge labels
—fw-bold	700	Headings H1-H4, toast titles, stat card labels
—fw-extrabold	800	Display text, section titles

Using Tokens in Code

Every token here is a CSS custom property in `frontend/src/styles/globals.css`. Colour, spacing, and radius tokens live in Tailwind's `@theme` block, so the token name is also the utility class. Motion tokens (duration, easing) stay plain CSS variables, referenced with arbitrary value syntax since Tailwind has no utility for those.

Guideline: prefer tokens over raw values

Raw hex values, pixel amounts, and millisecond literals should be avoided in component `className` strings when a token exists for the value. Colour, spacing, radius, shadow, and duration values should reference a token.

WRONG

```
className="bg-[#003A6B] p-4 rounded duration-150"
```

CORRECT

```
className="bg-brand-primary p-4 rounded-md duration-[--dur-fast]"
```

TOKEN	CLASS	CATEGORY
☐ <code>--brand-primary</code>	<code>bg-brand-primary</code>	Colour

TOKEN	CLASS	CATEGORY
☐ <code>—status-critical</code>	<code>bg-status-critical</code>	Colour
☐ <code>—color-text-secondary</code>	<code>text-color-text-secondary</code>	Colour
<code>—sp-4</code>	<code>p-4</code>	Spacing
<code>—sp-6</code>	<code>gap-6</code>	Spacing
<code>—dur-fast</code>	<code>duration-[—dur-fast]</code>	Motion
<code>—ease-standard</code>	<code>[transition-timing-function:var(—ease-standard)]</code>	Motion

Staying On-Brand with shadcn

shadcn's components (built on Radix UI) provide behaviour: focus trapping, keyboard navigation, ARIA wiring, positioning. They do not provide this project's visual design. Each one ships with its own default Tailwind classes for colour (`bg-primary`, `bg-background`, and so on). Those get replaced with this guide's own token classes when the component is built. The behavioural structure, `DialogContent`, `SelectTrigger`, `PopoverContent`, and so on, stays untouched.

shadcn's `Button` default variant ships as:

```
className="bg-primary text-primary-foreground hover:bg-primary/80"
```

It becomes:

```
className="bg-brand-primary text-color-text-inverse hover:bg-brand-primary/80"
```

Edit this in the component's own file, not at each usage site. shadcn's source is copied directly into `frontend/src/components/ui/*.tsx`, it is not an npm package, so editing it is expected and permanent, it will not be overwritten by anything except re-running the shadcn CLI for that specific component. Change the class string inside the component's own variant definition (for `Button`, that is `frontend/src/components/ui/button.tsx`), and every existing and future usage of `<Button variant="default">` picks it up automatically.

The same replacement applies to every component in the [Component Map](#): keep the primitive for its behaviour, replace its default classes with the tokens and custom CSS documented for that component in this guide. No variable aliasing, no separate file to keep in sync, the token names are what get written in the component itself. If a new shadcn component is added later via the CLI, it arrives with default classes again, apply this same replacement to it once, using whichever spec section of this guide covers it (or the Component Map row if it has no dedicated spec).

Colour System

Three tiers. Primary is the brand. Secondary covers the supporting palette, surfaces, and structural colours. Spot colours appear only when something needs communicating.

Primary

Brand Primary

—brand-primary

#003A6B

Navigation sidebar, hero backgrounds, primary filled buttons.

Brand Mid

—brand-mid

#174585

Interactive elements, links on light backgrounds.

Secondary

Brand Deep

—color-surface-deep

#002354

Dark decorative backgrounds. Also the hover/pressed state for text that defaults to Brand Primary.

Teal

—brand-teal

#056595

Decorative overlays, secondary icon backgrounds.

Muted

—brand-muted

#4F7392

Secondary labels at 24px+ (large text). Not for body copy on light.

Large text on light only

Light Blue

—brand-light

#8EADC4

Decorative on dark surfaces only. Not for body or UI text anywhere.

Decorative on dark only

Off White —color-surface-bg #F2F2F2 Page canvas. Cards and panels sit on top of it.	White —color-surface-raised #FFFFFF Cards, panels, modals. Raised one level above the Off White canvas.	Steel —brand-steel #6897B8 Decorative on dark surfaces, large text labels on dark. Not for body text on any background. Decorative on dark only	Ink —color-text-primary #313131 All body copy on light surfaces. 11.6:1 on surface-bg, 13.0:1 on white.
#FFF Inverse —color-text-inverse #FFFFFF All text on dark surfaces.	Light Grey —color-border #D8DAD6 Card borders, dividers, separators. Not intended for text. Used as a border for disable fields.	Input Border —color-input-border #767676 Input field outlines at rest. For UI components against white. Also used for disabled field text.	

Which secondary blue?

TOKEN	VALUE	WHERE TO USE IT	WHERE NOT TO USE IT
☐ —color-text-secondary (= —brand-muted)	#4F7392	Timestamps, metadata, zone codes, table secondary columns. On surface-raised at 14px+; on surface-bg at 24px+ only, except	Body copy, form hints, labels below 14px, any text

TOKEN	VALUE	WHERE TO USE IT	WHERE NOT TO USE IT
		on disabled/inactive controls, which are exempt from contrast requirements.	on surface-bg at small sizes.
☐ <code>—brand-steel</code>	#6897B8	Decorative strokes, dividers, and large display labels on dark surfaces.	Any text on light surfaces. Any text smaller than 24px on dark surfaces.
☐ <code>—brand-light</code>	#8EADC4	Purely decorative on dark surfaces. Icon strokes in subtle states, snippets on dark backgrounds.	UI text anywhere. Any context where it must be readable.

Spot

Spot colours appear only where meaning is needed. Status changes, data differentiation in diagrams, and chart series. Prefer not to use them for decoration.

The status subset carries risk level information. When a ranger sees green on the heatmap it must mean the zone is clear.

Navy

`—spot-navy`

#103364

Chart contrast, instructional diagrams and graphics.

Cobalt

`—spot-cobalt`

#2042B2

High visibility diagram elements, chart series.

Sky

`—spot-sky`

#0580BB

Chart series, diagram contrast alongside primary blues.

Blue

`—spot-blue`

#0070BF

Data differentiation in diagrams and charts.

Aqua

—spot-aqua

#009193

Map overlays, chart series,
diagram fills.

Safe

—status-safe

#06B050

Safe zones, patrol clear, data sync
success, form validation pass.

Caution

—status-caution

#FAE203

Low risk monitoring, non blocking
warnings. Use dark text on this fill.

Alert

—status-alert

#FF9300

Elevated risk, blocking validation
warnings. One step below critical.

Critical

—status-critical

#B30000

Immediate action required. Active
poaching incident, form errors,
destructive actions.

Status Text on Light

Darker variants of the status colours for text on light surfaces. These tokens exist only for text use. Prefer not to apply them to icon backgrounds or status fills.

Safe Text

—status-safe-text

#0A7D3A

Safe state text on light surfaces: toast titles, form validation pass messages.

Text on light only

Caution + Alert Text

—status-caution-text

#7A4800

Caution and alert state text on light surfaces: warning toast titles, elevated-risk stat values. Both yellow and orange are too light for text. This token covers both.

Text on light only

Critical Text

—status-critical-text

#9B0000

Critical state text on light surfaces: form error messages, error toast titles, destructive action labels.

Text on light only

Typography

One typeface family, two personalities. Barlow Condensed at 700 and 800 for all display and heading text. Barlow at 400, 500 and 600 for body and UI text.

DISPLAY

Barlow Condensed 800
—text-5xl (clamp 3.5–5.6rem)
line-height 0.95

Savanna Sentinel

HEADING 1

Barlow Condensed 700
—text-4xl (clamp 3.16–4.2rem)
line-height 1.05

Patrol Planner

HEADING 2

Barlow Condensed 700
—text-3xl (clamp 2.37–3.16rem)
line-height 1.1

Risk Heat Map

HEADING 3

Barlow Condensed 700
—text-2xl (2.369rem / 38px)
line-height 1.15

Report Received

HEADING 4

Barlow Condensed 700
—text-xl (1.778rem / 28px)
line-height 1.25

Ranger Field Update

BODY LARGE

Barlow 400

—text-lg (1.333rem / 21px)

line-height 1.65

Ranger logged a sighting in Zone B-4 at 04:32 SAST.

BODY

Barlow 400

—text-base (1rem / 16px)

line-height 1.65

Rangers use this platform to log sightings, review heatmaps, and track reported incidents across the reserve.

UI / LABEL

Barlow 500

—text-sm (0.875rem / 14px)

line-height 1.5

Submit Field Report

CAPTION

Barlow 500

—text-xs (0.75rem / 12px)

line-height 1.5

Zone A-3 / Updated 09:42 SAST / 3 high severity incidents

UI EMPHASIS

Barlow 600

—text-xs (0.75rem / 12px)

badges, column headers, overlines

PATROL ACTIVE · HIGH RISK · ZONE B-4

Spacing System

All spacing follows a 4pt base grid. Preferably use token values. Arbitrary pixel values break layout rhythm and make responsive adjustments unpredictable. Minimum touch target is 44x44 px for all interactive elements.

—sp-1	4px
—sp-2	8px
—sp-3	12px
—sp-4	16px
—sp-5	20px
—sp-6	24px
—sp-8	32px
—sp-10	40px
—sp-12	48px
—sp-16	64px
—sp-20	80px
—sp-24	96px

Breakpoints

The app uses four breakpoints from Tailwind v4 defaults. Default styles are mobile, desktop styles are additive via `md:` or `lg:` prefixes. `xl:` and `2xl:` are not used in this app. These are the target frontend's Tailwind breakpoints, not this static reference page's own layout, which uses its own `max-width` queries at 768px and 480px.

PREFIX	MIN-WIDTH	TYPICAL CONTEXT
(none)	0px	Mobile portrait, single column
sm:	640px	Mobile landscape, wider phones
md:	768px	Tablet, two-column layouts unlock, map sidebar appears
lg:	1024px	Desktop, full layout

BEHAVIOUR RULES

- The map sidebar (280px) appears at `md:` and above. Below `md:` it becomes a bottom drawer.
- Data tables gain horizontal scroll at narrow viewports. Wrap them in an `overflow-x-auto` container.
- The TopBar is the same height (56px) at all breakpoints.
- Stat card grids: 1 column by default, 2 columns at `sm:`, 4 columns at `md:`.

CODE EXAMPLE

```
Stat card grid
```

```
<div className="grid grid-cols-1 sm:grid-cols-2 md:grid-  
cols-4 gap-[--sp-4]">
```

Motion

Motion signals state changes. Duration and easing come from tokens, with one intentional exception: the skeleton loading shimmer runs on a hardcoded 1000ms interval that doesn't map to any duration token, since it's tuned to read as idle placeholder motion rather than a state transition. Wrap transform-driven motion that conveys entrance, exit, or spatial movement (panel slides, modal entrances, reveals) in `@media (prefers-reduced-motion: no-preference)`. Simple non-transform state transitions (colour, background, border, and opacity swaps on hover/focus) are not gated. Neither are small transform micro-transitions on compact UI affordances (a dropdown chevron's rotation, a tooltip's few pixel offset). Their motion is too brief and too small to register as a vestibular trigger.

Hover each track below to see the animation.

Fast

150ms / var(--ease-standard)



Hover states, focus rings, icon swaps, tooltip enter.

Normal

250ms / var(--ease-exit)



Modal and panel exit animations.

Deliberate

400ms / var(--ease-enter)



Page reveals, modal entrance, drawer and analysis panel entrance.

Enter

cubic-bezier(0, 0, 0.2, 1) / decelerates



Elements arriving. Modals, drawers, notifications.

Exit

cubic-bezier(0.4, 0, 1, 1) / accelerates



Elements leaving. Dismissed drawers, closed modals.

Performance: will-change

Apply `will-change: transform` to elements whose animations will play frequently when interacted with. This tells the browser to promote the element to its own compositor layer before the animation starts, which reduces jitter.

Do not apply it globally. Each promoted layer consumes GPU memory, which matters on the hardware rangers carry. Add it only where a repeated animation is confirmed, and remove it from elements that stop animating.

Iconography

Icons use Lucide React (the [lucide-react](#) package), defaulting to a 2px stroke weight at 16-24px. Every icon has a text label or an aria-label. Colour shows status, but it's never the only indicator. There's always a label or icon shape alongside it. (This static reference page can't run React, so its examples are rendered with Lucide's vanilla UMD build instead, at the same defaults.)

Examples are shown below.

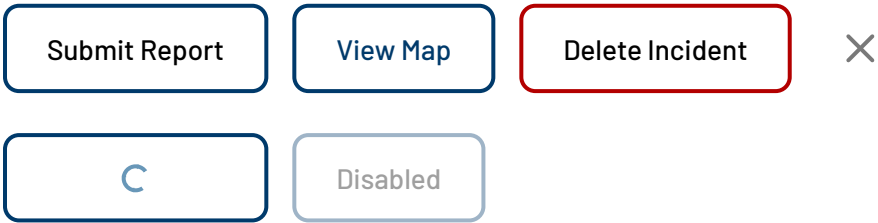


A few contexts deviate from some of the rules. For example, Empty-state and notification-empty icons run larger (24px and 32px) with a lighter 1.5px stroke to read as illustrative rather than functional. The file upload icon uses 1.5px, and the filter chip's remove icon is 12px at 2.5px stroke to stay legible at that size.

Components

Components are built on shadcn/ui for behaviour and accessibility. Visual styling comes from this guide's tokens, applied directly. Clear and distinguishable components are important for immediate recognition. The components shown here are examples to visualise what these components look like and how they generally behave. They do not represent the exact and/or only appearances.

Buttons



STATE	TRIGGER	VISUAL CHANGE
default	Rest	Full opacity, solid fill for primary
hover	Mouse over	Primary: opacity 87%. Danger: opacity 80%. Outline: background shifts to surface-bg. Ghost: opacity 100%, background shifts to surface-bg. All 150ms standard ease

STATE	TRIGGER	VISUAL CHANGE
active	Mouse/touch pressed	Inset shadow and tint colour overlay, text unaffected. Ghost: no press feedback, the action it triggers (dismiss, close) is immediate and obvious enough on its own
focus-visible	Keyboard navigation	2px solid outline, 2px offset: outline: 2px solid var(--brand-primary)
loading	Async action pending	Spinner overlay, pointer-events: none
disabled	Not available	Opacity 38%, pointer-events: none

Form Inputs

Zone designation

e.g. Zone A-3

Enter the patrol zone identifier.

GPS coordinates

-25.7461, 28.1

Coordinates appear incomplete. Include both latitude and longitude.

Disabled field

Not available offline

Navigation

Dashboard

Heatmap

Patrol Routes

Field Reports

Admin

Form Controls

SELECT / DROPDOWN

Reserve zone

Select a zone...

Default resting state.

Reserve zone (error)

Select a zone...

Please select a zone before submitting.

Reserve zone (disabled)

Not available offline

STATE	TRIGGER	VISUAL CHANGE
default	Rest	—color-input-border border, —radius-md corners, —sp-3 padding (right padding widened to —sp-8 to clear the chevron), chevron-down caret aligned right
hover	Mouse over	Border shifts to —brand-steel at 150ms standard ease (—dur-fast)
focus-visible	Keyboard or click	—brand-primary border, 2px outline offset 2px
error	Validation fail	—status-critical border, error message below

CHECKBOX

- ☐ Alert notifications
- ☒ Email summaries (checked)
- ☐ SMS alerts (disabled)
- ☐ Weekly digest (checked, disabled)
- ☐ Select all zones (indeterminate)
- ☐ Accept terms (error, required)

STATE	VISUAL CHANGE
default	16x16px, 2px border-radius, 2px —color-border border, clear fill
hover	Border shifts to —brand-steel
focus-visible	2px —brand-primary outline, offset 2px
checked	—brand-primary fill and border, 6x6px white square indicator (1px border-radius)
indeterminate	—brand-primary fill, white horizontal bar (8x2px)
error	—status-critical border
disabled (unchecked)	—color-border border, —color-surface-bg fill, cursor not-allowed

	Full	Below
disabled	Not available	—color-surface-bg background, —color-input-border text, cursor not-allowed

TEXTAREA

Incident description

Describe what you observed...

0 / 300

STATE	TRIGGER	VISUAL CHANGE
default	Rest	Same border, padding, and radius as text input. Min-height 96px, resize: vertical only
focus-visible	Keyboard or click	—brand-primary border, 2px outline offset 2px
error	Validation fail	—status-critical border, error message below
disabled	Not available	—color-surface-bg background, —color-input-border text, cursor not-allowed

disabled	—color-border fill, —color-text-secondary
(checked)	indicator, cursor not-allowed

MULTI-SELECT DROPDOWN

A checkbox list behind a single trigger button, for choosing several values from one field (e.g. filtering reports by more than one ranger).

The trigger's summary text reflects selection count: "None Selected" when empty, the single item's label when exactly one is checked, and "X selected" once two or more are checked.

Ranger

None Selected

Ranger

J. Mthembu

Ranger

3 selected

ELEMENT / STATE	CLASS	VISUAL CHANGE
Wrapper	.form-multiselect	Plain grouping element; carries no layout of its own
Trigger, default	.form-multiselect-trigger	Identical footprint to .form-input: 1.5px —color-input-border border, —radius-md corners, 44px min-height, chevron icon at right —sp-3

character count	maxlength set	Count displayed below-right in —text-xs —color-text-primary. Updates live on input
------------------------	------------------	---

TOGGLE SWITCH

- ☐ Layer visibility (off)
- ☒ Push notifications (on)
- ☐ SMS alerts (disabled)

STATE	VISUAL CHANGE
off	40x20px pill, —color-border track, 16x16px white thumb at left (2px inset), —shadow-sm
on	—brand-primary track, thumb translated 20px right in —dur-fast ease-standard
hover	Track darkens toward —brand-steel when off, or dims to 87% opacity when on
focus-visible	2px —brand-primary outline on track, offset 2px
disabled	Opacity 38%, cursor not-allowed on track and label

RADIO

Trigger, hover	.form-multiselect-trigger:hover	Border shifts to —brand-steel
Trigger, focus-visible	.form-multiselect-trigger:focus-visible	—brand-primary border, 2px outline offset 2px
Trigger, open	.form-multiselect-trigger.is-open	—brand-primary border; chevron rotates 180deg via the ::after pseudo-element
Trigger, disabled	.form-multiselect-trigger:disabled	—color-surface-bg background, —color-input-border text, cursor not-allowed
Summary, none selected	.form-multiselect-summary.is-placeholder	Text reads "None Selected"; —color-input-border colour, same muted treatment as .form-input::placeholder
Summary, one selected	.form-multiselect-summary	Text reads the single checked option's label (e.g. "J. Mthembu"), —color-text-primary colour
Summary, two or more selected	.form-multiselect-summary	Text reads "X selected" where X is the checked count, —color-text-primary

Patrol frequency

- ☐ Daily
- ☒ Weekly (selected)
- ☐ Monthly (disabled)

STATE	VISUAL CHANGE
default	20x20px circle, —color-input-border border, clear fill
hover	Border shifts to —brand-steel
focus-visible	2px —brand-primary outline, offset 2px
selected	Outer ring —brand-primary (5px border), inner white fill 10px diameter
disabled	—color-border ring, opacity 50%, cursor not-allowed

SLIDER

Used for continuous numeric selection. Time range, opacity, and intensity controls. Built with divs, not a native range input, so that the track, fill, and thumb can be fully styled.

SNAPSHOT SLIDER (MAP SIDEBAR)



more
selected

multiselect-
summary

x is the checked count; never
lists the individual labels

Dropdown panel

```
.form-  
multiselect-  
panel
```

position: fixed; top/left/width computed from the trigger's bounding rect via JS on open and on scroll/resize, so it floats independently of any scrollable ancestor instead of being clipped or resizing it; — color-surface-raised background; —shadow-md; opens downward with a 4px gap

Option row

```
.form-  
multiselect-  
option
```

Extends `.form-check-wrapper`;
full-width, 44px min-height, —
radius-sm hover fill; checking
a box does not close the panel

DATE / TIME PICKER

Date and time inputs reuse the `.form-input` class. Native browser pickers open on click. For a calendar popover (richer UX), use `shadcn Calendar` inside a `Popover` triggered by a styled button.

Incident date

2026 / 05 / 14 

Snapshot: 15 Dec 2025

OPACITY SLIDER (20%)

Heatmap opacity 20%

DISABLED

Patrol intensity 60%

PART	SPEC
------	------

track 6px tall, —color-border background, border-radius 3px, position relative

Stat Cards

fill Same height as track, left-anchored, —brand-primary background. Width mirrors the current

PATROL COVERAGE

73%

Up from 68% last week

4 zones - 10:23 SAST

FIELD REPORTS TODAY

7

No reports from Zone C since 08:15

Zones B-C - 10:24 SAST

OPEN INCIDENTS

14

2 reported in the last 6 hours

38%, cursor not-allowed.

Patrol start time

06 : 30

Sighting logged at

2026 / 05 / 14 , 06 : 30

End date (unavailable)

2026 / 07 / 31

STATE	TRIGGER	VISUAL CHANGE
-------	---------	---------------

Feedback Toasts

Identical to text input: 1.5px —color-input-border border, —radius-md corners, 44px min-height

default	Rest	
---------	------	--

In the app, `.toast-stack` is fixed to the viewport corner (`top-right`). This demo is confined to the box below so it doesn't cover the page.

focus-	Keyboard	—brand-primary border, 2px outline
--------	----------	------------------------------------

Report saved

Field report #1042 was saved and synced to the server.

10:23 SAST

Working offline

Your reports are queued and will sync when you reconnect.

10:24 SAST

Top Bar

6 zones - 10:31 SAST

The persistent header above every protected page. Height is 56px. Background always surface-raised. Page title lives in the page content area, not the top bar

Couldn't sync

Server unreachable. Check your connection and try again.

10:31 SAST

ONLINE STATE

Savanna Sentinel

JM

OFFLINE STATE

Savanna Sentinel

JM

SLOT	CONTENT	RULE
Left	Burger menu button + wordmark	Always visible. Tapping the burger opens the side navigation drawer. Wordmark hidden at viewports narrower than 480px.
Right (online)	Notifications, Help, Avatar	Icon buttons for global actions. Avatar links to the account page. The Notifications button shows a dot if an unread notification is present.
Right (offline)	Same as above with Globe-off added	The <code>globe-off</code> Lucide icon appears to the left of Notifications when the device has no internet connection. Colour: <code>--brand-primary</code> . Clicking it opens a small "No Internet connection" popover (Escape or an outside click closes it).

Offline tone: since this is a PWA, being offline is expected. `--brand-primary` is used for the icon, not a status colour. The "Working offline" toast appears only when the

user submits a report that enters the sync queue. The icon is a passive background indicator.

Side navigation drawer: Tapping the burger opens a full height drawer that slides in from the left and overlays page content. It contains the primary navigation links filtered by the user's role, the app wordmark, and a sign out action at the bottom. The drawer closes when a nav item is selected or when the user taps outside it.

Badges and Status Chips

Inline status tags for user state, incident severity, and zone classification, etc. The border colour encodes the status level. Never use a badge as the only mechanism to communicate critical status, pair with a label / icon.



VARIANT	CLASS	EXAMPLE USAGE
Active	<code>.badge-safe</code>	User active, zone clear, sync success
Pending	<code>.badge-caution</code>	Awaiting review, low-risk monitoring
Elevated	<code>.badge-alert</code>	Elevated risk, requires attention
Critical	<code>.badge-critical</code>	Active incident, user suspended, form error
Inactive	<code>.badge-neutral</code>	Inactive user, classification label, zone tag

Data Table

Used in field reports, user admin, and audit logs. Header uses brand-primary so severity and status badges remain readable against the dark strip. Secondary data (timestamps) uses `--color-text-secondary` at text-sm to reduce visual weight without losing information.

REPORT	ZONE	SEVERITY	RANGER	TIME
#1042	Zone A-3	Critical	J. Mokoena	10:31 SAST
#1041	Zone B-1	Elevated	S. van Niekerk	09:44 SAST
#1040	Zone C-2	Cleared	T. Dlamini	08:19 SAST

Modal / Dialog

Use shadcn's [Dialog](#) or [AlertDialog](#). The overlay is rendered by [DialogOverlay](#) (Tailwind [bg-black/80](#)). Enter at 400ms ease-enter, exit at 250ms ease-exit. Do not close on backdrop click for destructive actions.

Live demo: opens as a real overlay. This one is a destructive confirmation, so backdrop click won't dismiss it (Escape and the buttons still do)

[Preview delete dialog](#)

Empty State

Shown when a list has no content. An offline-first app will show empty states frequently during first sync. Icon uses muted stroke, title uses brand-primary, body uses text-primary. Include a primary action only when the user can take immediate action.



No reports yet

Field reports submitted in your zones
will appear here.

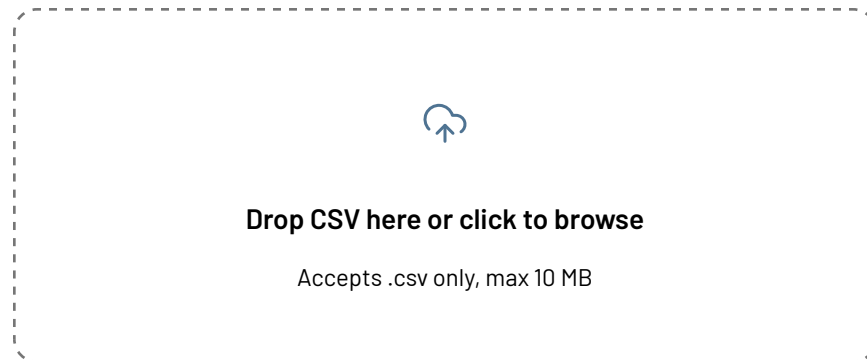
[Submit First Report](#)

File Upload

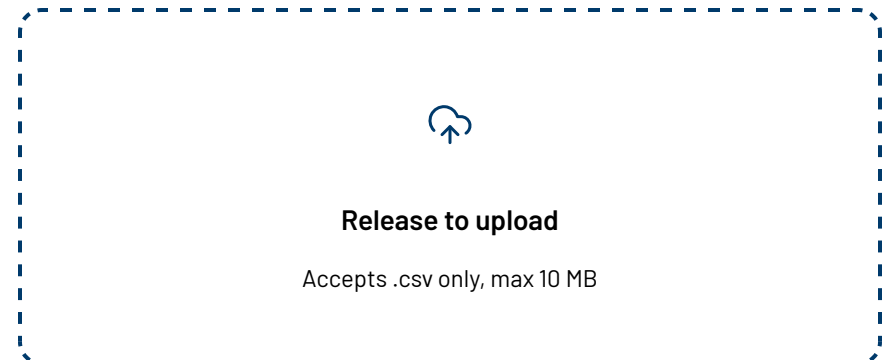
A dashed border drop zone accepts files. Drag-and-drop is supported (not implemented on this page). Clicking also triggers the native file picker. The constraint

label (file type and size limit) is shown at idle and drag-over. Uploading replaces it with progress/file-size info, success replaces it with the uploaded file's name and size, and error replaces it with the failure reason. States: idle, drag-over, uploading (with progress bar), success, and error.

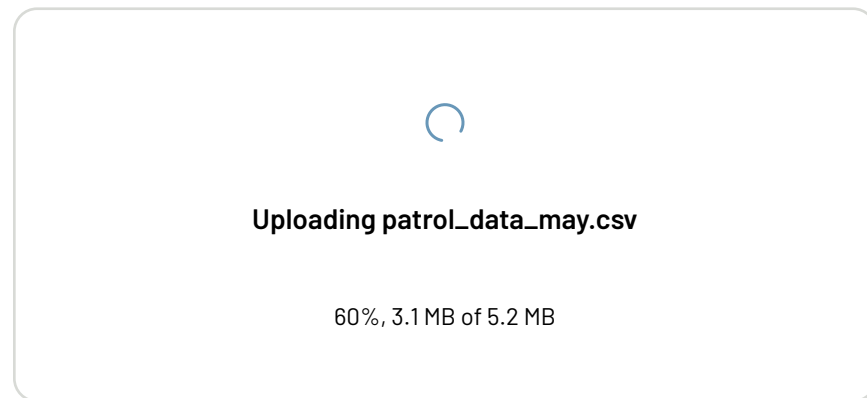
Idle



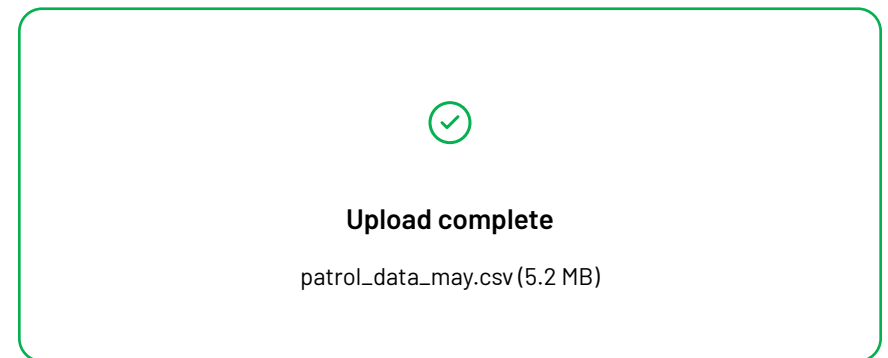
Drag over



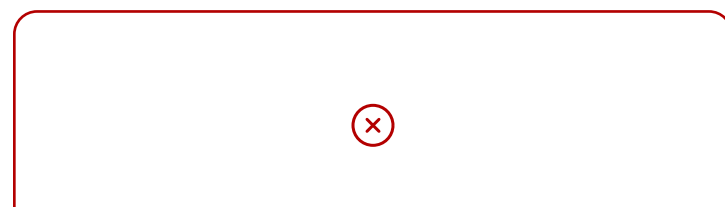
Uploading

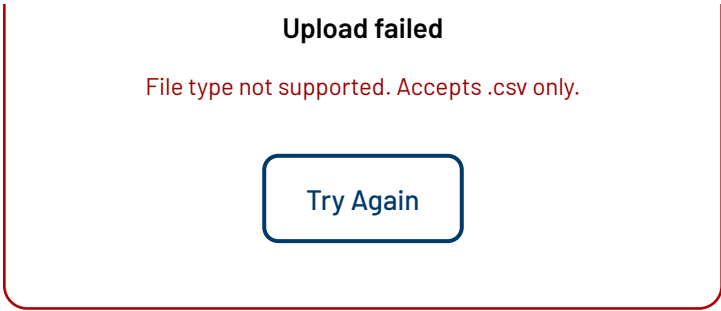


Success



Error





STATE	BORDER	ICON	NOTES
Idle	1.5px dashed —color-input-border	upload-cloud, —brand-muted	Default. Listens for dragenter and click
Drag over	2px dashed —brand-primary	upload-cloud, —brand-primary	Background: —brand-primary at 5% over surface-raised
Uploading	1.5px solid —color-border	loader-circle, —brand-steel; spins	Title shows "Uploading" + filename. Hint shows percent and byte progress alongside the progress bar.
Success	1.5px solid —status-safe	circle-check, —status-safe	Background: —status-safe at 8% over surface-raised
Error	1.5px solid —status-critical	circle-x, —status-critical	Background: —status-critical at 6%; hint shows reason. Try Again button visible

Pagination

Previous and next buttons sit at the edges, numbered page chips fill the middle. The

current page chip uses filled brand styling. Boundary buttons are disabled when the first or last page is active. Ellipsis spans mark gaps in the page sequence.

Middle page (4 of 7)



First page (previous disabled)



Last page (next disabled)



ELEMENT	CLASS	NOTES
Container	.pagination	nav element; flex row; gap —sp-1; flex-wrap for narrow viewports
Page button	.pagination-btn	44px min touch target; 1.5px border; —color-border; radius-sm
Active page	.pagination-btn.active	—brand-primary background; inverse text; aria-current="page"
Disabled boundary	.pagination-btn:disabled	35% opacity; cursor not-allowed; applied to prev on page 1 and next on last page
Ellipsis	.pagination-ellipsis	Non-interactive span; aria-hidden="true"; —color-text-primary

Search and Filter Bar

A search input with a leading Lucide search icon sits beside a filter button. Once the field has a value, a circular clear button appears at its trailing edge to reset the search. Clicking the filter button opens the filter panel: a dropdown panel on desktop, anchored below the button, and a bottom drawer below the md breakpoint. When filters are active, a chip row below the bar shows each applied filter with a remove control, and the filter button displays a count badge.

No active filters

 Search reports...

 Filters

Active filters applied

 elephant

×

 Filters 2

Status: Open ×

Status: Under Review ×

Filter panel (opens from the Filters button)

Filters

STATUS

Under Review

RANGER

J. Mthembu

DATE RANGE

From

yyyy / mm / dd 

To

yyyy / mm / dd 

Clear

Apply

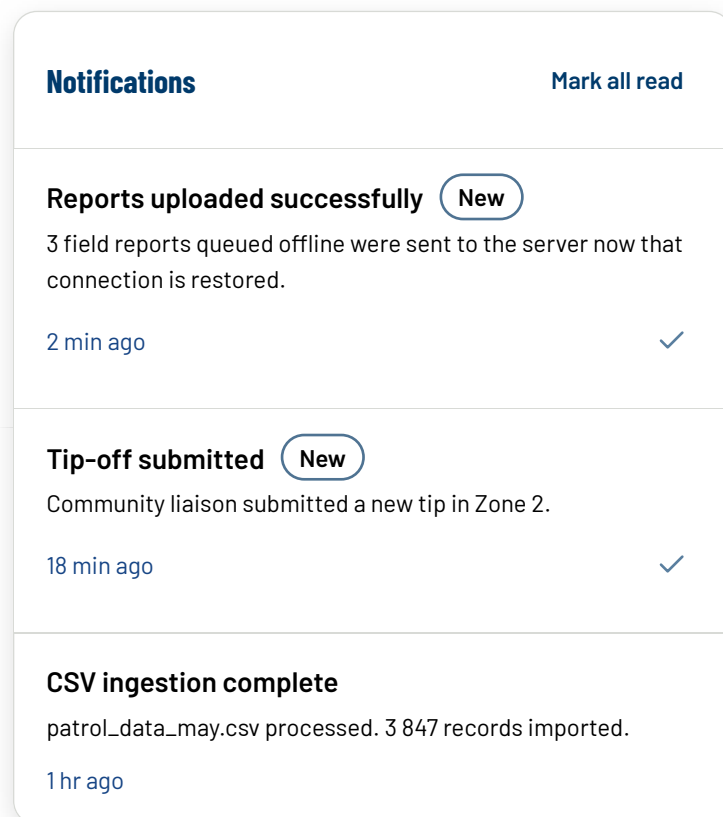
ELEMENT	CLASS	NOTES
Bar wrapper	.search-bar	flex row; gap —sp-2; full width
Input wrapper	.search-input-wrap	position relative; flex: 1; contains icon and input
Search input	.form-input.search-input	Extends .form-input; padding-left/right: —sp-10 to clear the search icon and clear button, each with a —sp-3 gap
Search icon	.search-icon	Absolute; left —sp-3; vertically centred; 16px stroke; —brand-muted; pointer-events: none
Search clear	.search-clear-btn	Absolute; right —sp-3; 20px circle, —color-border fill; same hover/focus/hit-area treatment as .filter-

ELEMENT	CLASS	NOTES
button		chip-remove; rendered only when the field has a value
Filter button	.btn.btn-outline.filter-btn	Flex; gap —sp-2; shrinks to content; holds optional count badge
Count badge	.filter-count-badge	—brand-primary bg; inverse text; pill; 20px min size; shown when active filter count is greater than 0
Chip row	.filter-chip-row	flex-wrap row; gap —sp-2; rendered below .search-bar when filters are active
Filter chip	.filter-chip	Inline-flex pill (radius 99px), matching the badge family; border —color-border; height set by content, not a forced min-height
Chip remove	.filter-chip-remove	20px circular button; hover fills —status-critical. An absolutely-positioned ::before with inset: -12px expands the hit area to 44x44px without enlarging the visible chip
Filter panel	.filter-panel	Popover opened by the filter button; —color-surface-raised background; —shadow-md; max-width 320px; bottom drawer below the md breakpoint
Panel header	.filter-panel-header	Border-bottom —color-border; holds the panel title
Filter group	.filter-group	One labelled control per group (multi-select dropdown or paired date inputs); —sp-5 gap between groups
Panel footer	.filter-panel-footer	Border-top —color-border; right-aligned Clear (btn-outline) and Apply (btn-primary), same order as the Modal footer

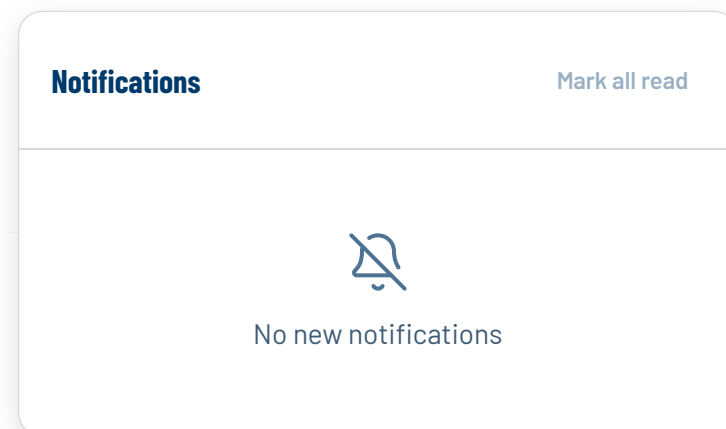
Notification Panel

Opened by the bell icon in the TopBar. Renders as a dropdown panel on desktop and a full-width bottom drawer anchored to the bottom edge of the viewport on smaller screens. Each item shows a title, body text, and a relative timestamp. Unread items also carry a "New" badge. A per-item check button marks that single notification as read, and the "Mark all read" action clears every unread item at once.

Panel with unread items



Empty state



ELEMENT

CLASS

NOTES

ELEMENT	CLASS	NOTES
Panel	.notif-panel	max-width 360px; surface-raised bg; shadow-md; radius-lg; border; z-index —z-dropdown
Panel, mobile (<768px)	.notif-panel (media query)	Becomes a bottom drawer: fixed to the viewport's bottom edge, full width, top corners only (radius-lg), no bottom border. Slides up on open (—dur-deliberate, —ease-enter). Appears instantly when motion is reduced.
Header	.notif-panel-header	flex; space-between; border-bottom; padding —sp-3 —sp-4
Title	.notif-panel-title	Barlow Condensed 700; —text-base; —brand-primary
Mark all read	.notif-mark-all	Ghost button; —text-xs; —brand-primary (11.5:1 on surface-raised). Disabled and dimmed when no unread items remain
List	.notif-list	ul; list-style none; max-height 400px; overflow-y auto
Item	.notif-item	padding —sp-3 —sp-4; border-bottom
Item (unread)	.notif-item.unread	Background: —color-surface-raised. Unread is marked by the "New" badge and the button.
Item header	.notif-item-header	flex; wrap; gap —sp-2; holds the title and, when unread, the "New" badge
Item footer	.notif-item-footer	flex; space-between; holds the timestamp and, when unread, the "mark as read" button
Timestamp	.notif-time	—text-xs; —brand-mid

ELEMENT	CLASS	NOTES
Mark as read	.notif-mark-read	44×44px icon button; check icon; —brand-primary; shown only on unread items
Empty state	.notif-empty	Flex column; centred; —brand-muted icon (32px); —text-sm caption; padding —sp-10 (top/bottom) —sp-6 (left/right)

Skeleton States

Skeletons replace content while async data loads. Each skeleton matches the general shape of the content it stands in for, so the user can see what's loading. A shimmer sweeps left to right. On reduced-motion systems the shimmer is replaced by a slow opacity pulse.

Text lines

Card skeleton

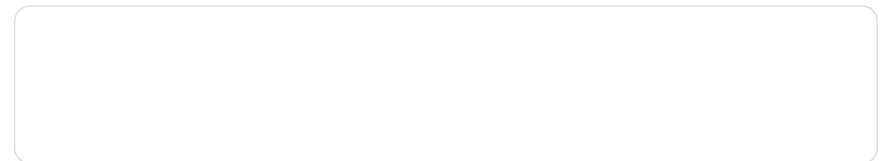


Table row skeletons (header always renders normally)

REPORT	ZONE	SEVERITY	TIME

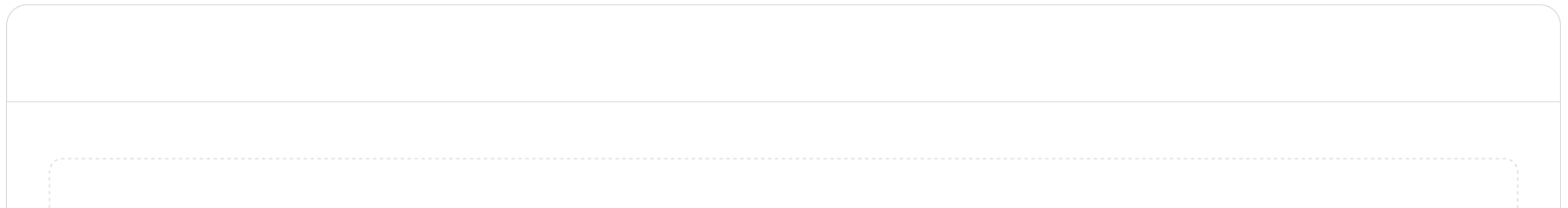
VARIANT	HEIGHT	BORDER RADIUS	USED FOR
Text line (text-xs)	12px	radius-sm (4px)	Secondary labels, timestamps
Text line (text-sm)	14px	radius-sm (4px)	Body text, table cells
Card	Full card height	radius-md (8px) on the outer wrapper; the <code>.skeleton</code> bars inside stay radius-sm (4px)	Route comparison cards, stat cards
Table row	Normal row height	radius-sm (4px)	Report lists, user tables
Map canvas	Not used	N/A	Base tiles load immediately; cells fade in

Line width pattern: avoid uniform widths across multiple sequential lines.

Standard Page Layout

All non-map pages use a centred content wrapper.

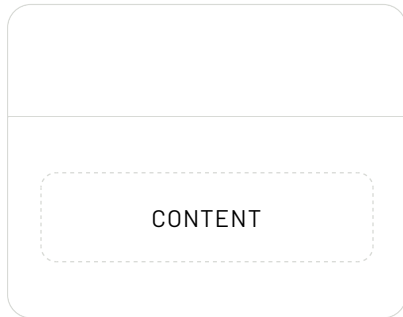
DESKTOP (MIN-WIDTH: 768PX)



CONTENT WRAPPER (MAX-WIDTH 1120PX, PADDING-X --SP-6, PADDING-TOP --SP-8)

MOBILE (BELOW 768PX)

Full width, padding-x —sp-4



BREAKPOINT	MAX-WIDTH	HORIZONTAL PADDING	NAVIGATION
Desktop (md: and above)	1120px, centred	—sp-6 (24px)	Burger in TopBar opens side drawer. No inline sidebar.
Mobile (below md:)	Full width	—sp-4 (16px)	Same burger. Tables scroll horizontally within their bordered wrapper rather than reflowing. Touch targets: 44x44px minimum.

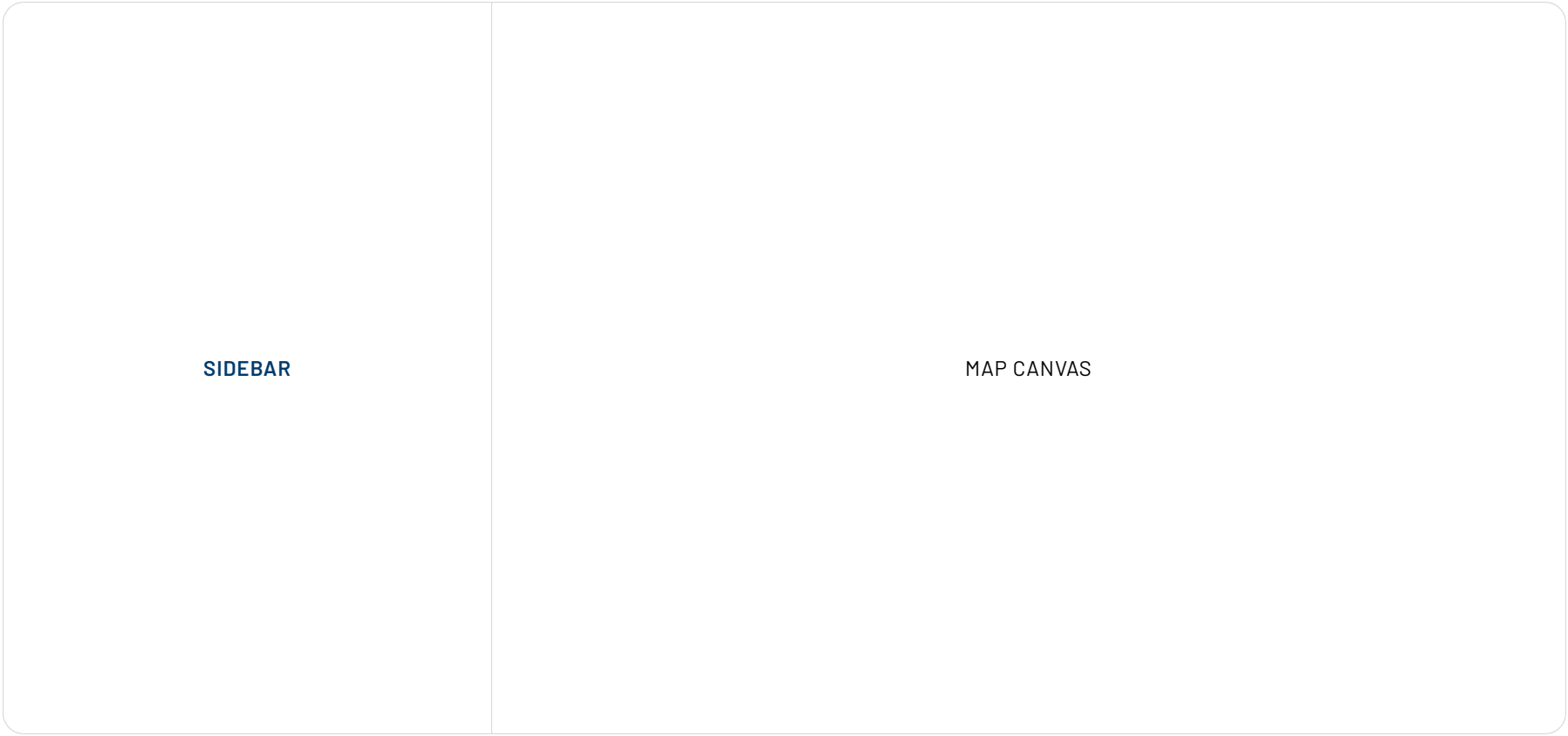
Map Components

Two pages use a full viewport map canvas: the Heatmap, and the Patrol Planner. Both share the general map components defined here. Each adds its own sidebar controls.

Page Layout

Map pages fill the full viewport height below the 56px TopBar. There is no max-width wrapper and no page padding. On desktop the left sidebar is 280px wide and the canvas fills the remaining space. On mobile the sidebar disappears and its content moves into a bottom drawer opened by a slide pill.

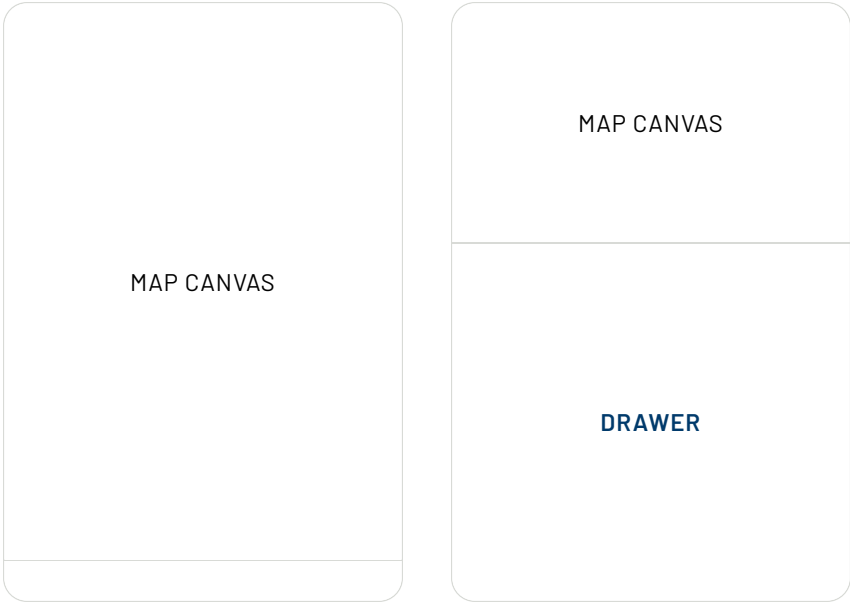
DESKTOP (MIN-WIDTH: 768PX)



MOBILE (BELOW 768PX)

Drawer closed

Drawer open



ZONE	DESKTOP	MOBILE (BELOW 768PX)
Sidebar	280px wide, full canvas height, scrollable. White background, right border. Contains page specific controls.	Hidden. Content moves to the bottom drawer.
Map canvas	flex: 1, fills remaining width and full height. Must have an explicit pixel height (not min-height) so MapLibre GL sizes correctly.	Fills remaining height above the bottom drawer.
Bottom drawer	Not present.	Always visible as a peek strip at the bottom. Swipe up to expand to 60% viewport height. White background, top corners rounded (radius-lg). Centred 32x4px drag handle in border colour. Dismissed by swiping down or tapping the backdrop.

Risk Colour Scale

The API returns a `risk_score` float between 0 and 1 for each grid cell. That score maps to one of four risk levels using evenly split thresholds. Thresholds must be stored as a single named constant in code so they can be changed without editing multiple files.

Low	Medium	High	Critical
0.00-0.25	0.25-0.50	0.50-0.75	0.75-1.00

PROPERTY	VALUE
Fill opacity (default)	0.55, user adjustable on the Heatmap page. Fixed at 0.30 on the Patrol Planner page.
Cell border	<code>0.5px solid rgba(255,255,255,0.25)</code>
Selected cell border	<code>2px solid rgba(255,255,255,1.0)</code>
Selected cell brightness	<code>filter: brightness(1.15)</code>

Interior Suppression Rule

Low-risk cells that are fully surrounded by other Low-risk cells (or by the grid boundary) do not render their green fill. Only Low cells that share a border with a

Medium, High, or Critical neighbour are shown, marking the edge of the safe zone. This prevents the safe zone from being painted solid green. All cells, including suppressed ones, remain clickable and open the popup.

Interior cell: 15%
opacity

Boundary cell: 55%
opacity

CONDITION	FILL OPACITY	CLICKABLE
Low cell, all 8 neighbours also Low (or absent at grid edge)	15%	Yes
Low cell, at least one neighbour is Medium, High, or Critical	55% (default)	Yes
Medium, High, or Critical cell	55% (default)	Yes

The 8-neighbour check can't be expressed as a MapLibre paint expression on its own, since a feature can't inspect its neighbours' properties at render time. The API precomputes an `is_interior` boolean per cell and includes it in the GeoJSON. The paint expression (see the Risk Colour Scale section above) only branches on that flag.

Loading State

While heatmap cell data is loading from the API, the base map tiles render immediately. A small pill floats in the top left corner of the canvas. When data arrives, cells fade in. The canvas is never blanked or skeletonised.

PROPERTY	VALUE
Position	Top-left of canvas, 8px from top and left edges
Spinner colour	—brand-steel, 16px diameter
Cell fade-in	opacity 0 to 1 over 250ms, standard ease cubic-bezier(0.2,0,0,1): individual cell fade-ins are small in-place updates, not a panel or overlay entrance, so they use the snappier standard curve rather than ease-enter
Pill disappears	When all cells have loaded

Map Controls

Three icon buttons float over the top-right corner of the canvas. They use the outline button style (`.btn-outline`) so they remain legible over any map tile. All buttons meet the 44x44px minimum touch target for field use with gloves.



PROPERTY	VALUE
Position	Top-right of canvas, 8px from top and right edges
Z-index	<code>var(--z-sticky) = 200</code>
Touch target	44x44px minimum
Gap (Zoom In / Zoom Out)	4px
Top margin (Reset North)	4px additional, separating it from the zoom pair
Button style	Secondary outline with box-shadow: <code>var(--shadow-sm)</code>

Risk Legend

The risk legend floats in the bottom-right corner of the canvas on both the Heatmap and Patrol Planner pages. On the Patrol Planner page it defaults to the collapsed pill because the heatmap is background context, not the primary view.

Full panel (Heatmap page default)

RISK LEVEL

☐

Critical

☐

High

☐

Medium

☐

Low

Collapsed pill (mobile / Patrol Planner default)

C

H

M

L

PROPERTY	VALUE
Position	Bottom-right of canvas, 8px from bottom and right edges
Heatmap page default	Full panel on desktop. Collapsed pill on mobile, pinned 12px above the bottom drawer peek strip.
Patrol Planner default	Collapsed pill on both desktop and mobile. Tap to expand.
Swatch size	16x16px, border-radius: 2px, border: 1px solid rgba(216,218,214,0.5)

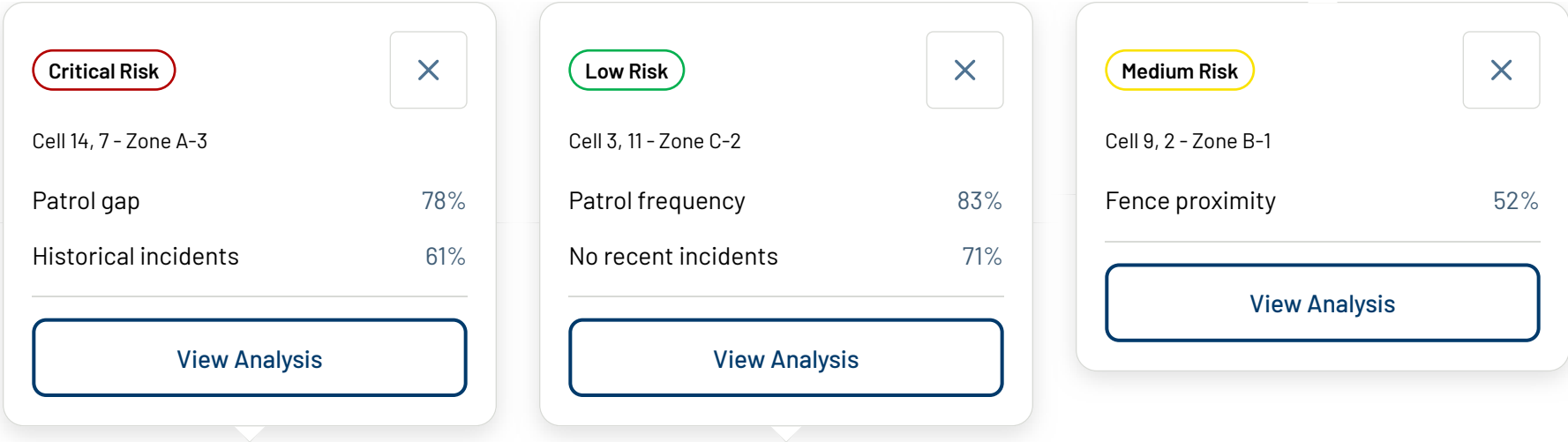
Cell Popup

Clicking or tapping any heatmap cell on either page opens a popup with the cell's risk level and top contributing factors. The popup has a downward-pointing tail centred on the selected cell. On mobile the popup anchors to the top of the canvas rather than the cell.

Critical cell

Low cell

Repositioned near canvas bottom (tail flipped)



PROPERTY	VALUE
Width	280px
Desktop position	Anchored to the geographic centre of the clicked cell. Tracks the cell as the user pans or zooms the map. Repositioned automatically if it would overflow the canvas edge.
Mobile position	Anchored to the top of the canvas
Open animation	Scale 95% to 100% over 150ms, cubic-bezier(0,0,0.2,1)
Dismissed by	Close button, Escape key, or clicking outside the popup
Multiple cells	Clicking a different cell closes the current popup and opens a new one
Confidence bar height	4px

PROPERTY	VALUE
Tail	Downward-pointing triangle at the horizontal centre of the popup bottom, pointing at the clicked cell. Flips to upward (<code>.tail-up</code>) when the popup is repositioned below the cell to avoid canvas overflow.

Advanced Analysis Panel

Clicking View Analysis in the cell popup closes the popup and slides in a detail panel from the right edge of the canvas. It shows the full factor breakdown and model metadata. The panel overlays the map, the canvas does not shrink.

Live demo: the panel overlays the canvas without resizing it. Close it to see the slide out, then reopen to see it slide back in

Zone Analysis

Cell 14, 7 - Zone A-3

Critical Risk

Risk score: 82%

CONTRIBUTING FACTORS

Patrol gap

78%

Historical incidents

61%

Proximity to water

44%

Model

Random Forest v2.1

F1 Score

0.87

Last trained

14 May 2026

Reopen analysis panel

PROPERTY	VALUE
Desktop width	320px, slides in from the right canvas edge
Mobile	Full canvas width, slides in from the right canvas edge
Overlay behavior	Overlays the map canvas. Canvas does not resize.
Open animation	Slides in from right over 400ms, cubic-bezier(0,0,0.2,1)
Close animation	Slides out to right over 250ms, cubic-bezier(0.4,0,1,1)
Confidence bar height	6px (taller than popup's 4px). Use .confidence-bar-track.tall
Factor rows	All factors shown, ordered highest to lowest confidence.
Footer	"Close Analysis" btn-outline, full width, sticks to panel bottom

Heatmap Sidebar

On the Heatmap page, a left sidebar (desktop) or bottom drawer (mobile) is used for manipulating what data is shown on the map.

TIME RANGE

Jan 2025May 2026



Snapshot: 14 May 2026

14 May 2026

LAYERS

☒ Risk Heatmap

☐ Roads

☐ Incidents

☐ Water Sources

☐ Fence Lines

Heatmap Opacity55%



SUMMARY

Critical cells6

High-risk cells23

Incidents (30d)142

Last updated14 May

BLOCK

CONTENT

RULE

BLOCK	CONTENT	RULE
Time Range	Date boundary labels + range slider + snapshot date + snapshot select	Filled track: brand-primary. Unfilled track: color-border. Thumb: white fill with a 2px brand-primary border. Select below the slider lists only available snapshot dates. Selecting a date moves the slider to the matching position.
Layers	Example checkbox rows: Risk Heatmap, Roads, Incidents, Water Sources, Fence Lines	Only Risk Heatmap is active by default.
Heatmap Opacity	Range slider 0%-100%, step 1%, default 55%. Current value shown inline.	Only interactive when Risk Heatmap layer is on. Interior-suppressed Low cells cannot go above 15% opacity.
Summary	Critical cells, High-risk cells, Incidents (30d), Last updated	Critical cells value: status-critical colour. High-risk cells value: status-caution colour.

Patrol Planner Sidebar

The Patrol Planner uses the same full viewport canvas as the Heatmap page. Heatmap cells are still visible at fixed 30% opacity and remain clickable. The sidebar contains constraint inputs and the generated route alternatives.

Plan block + skeleton loading state

Alternatives block after routes load

PLAN ROUTE

Start Point

Click map or enter coordinates

Max Time

0

min

Max Fuel

0

L

Generate Routes

ALTERNATIVES

ALTERNATIVES

Route A

Selected

Est. Time

38 min

Est. Fuel

16 L

Risk Coverage

74%

Route B

Select

Est. Time

45 min

Est. Fuel

21 L

Risk Coverage

58%

Route C

Select

Est. Time

52 min

Est. Fuel

25 L

Risk Coverage

31%

Route generation status pill (shown on canvas while job runs)

Planning route...

PROPERTY	VALUE
Route A colour	—spot-sky
Route B colour	—spot-aqua
Route C colour	—spot-cobalt
Selected route line	5px, 100% opacity, —spot-navy
Unselected route lines	2px, 40% opacity, —spot-blue
Start marker	 Circle, brand-primary fill, 2px white border. Play icon in white.
End marker	 Diamond (square rotated 45deg), brand-primary fill, 2px white border. Square icon in white.
Marker z-order	Always above route lines, always visible regardless of selection
Heatmap opacity on this page	Fixed 0.30, no opacity slider
Cell interactivity on this page	Cells remain clickable. Same popup as the Heatmap page.
Risk Coverage colour thresholds	Below 40%: status-critical. 40-70%: status-caution. Above 70%: status-safe.
Default selection	Route A auto-selected when cards first appear

shadcn Component Map

Certain design pattern maps to a specific shadcn primitive. Use this table to find the correct component and variant before building.

DESIGN PATTERN	SHADCN COMPONENT	KEY PROPS / VARIANTS	NOTES
Primary button	Button	<code>variant="default"</code>	Replace <code>bg-primary</code> with <code>bg-brand-primary</code> directly on the component
Outline button	Button	<code>variant="outline"</code>	
Danger button	Button	<code>variant="destructive"</code>	
Ghost / icon button	Button	<code>variant="ghost", size="icon"</code>	Used for Top Bar actions, modal dismiss, notification actions
Badge	Badge	<code>variant</code> maps to status level	Override colours with token CSS. See Badges section
Toast	Toaster + <code>toast()</code>	<code>toast.success()</code> to <code>.toast-safe</code> , <code>toast.warning()</code> to <code>.toast-</code>	Uses sonner. Sonner's default method names do not match this guide's severity vocabulary,

DESIGN PATTERN	SHADCN COMPONENT	KEY PROPS / VARIANTS	NOTES
		<code>caution</code> , <code>toast.error()</code> to <code>.toast-critical</code>	apply the class per the mapping above. Match title/body to voice/tone section
Modal (standard)	Dialog	DialogContent, DialogHeader, DialogFooter	
Modal (destructive)	AlertDialog		Do not close on backdrop click. Requires explicit cancel or confirm
Side nav drawer	Sheet	<code>side="left"</code>	Full-height overlay, z-index <code>--z-modal</code>
Dropdown panel (responsive)	Popover + Sheet	<code>Popover</code> at md breakpoint and up, <code>Sheet side="bottom"</code> below md	Used by the Notification Panel and the filter panel in Search and Filter Bar, anchored below the trigger button on desktop, full-width bottom drawer below md
Map sidebar bottom drawer	Drawer (vault)	Peek strip + drag handle, snap to 60% viewport height	Used by the Heatmap Sidebar and Patrol Planner Sidebar on mobile (see Page Layout spec). Needs swipe-to-expand and swipe-down-to-dismiss gestures with a persistent peek state, which <code>Sheet</code> does not provide, use <code>Drawer</code> instead
Advanced Analysis Panel	Sheet	<code>side="right"</code>	Overlays the map canvas without resizing it, same behaviour as the side nav drawer, mirrored to the right edge

DESIGN PATTERN	SHADCN COMPONENT	KEY PROPS / VARIANTS	NOTES
Skeleton	Skeleton	Pass width and height inline	Heights per skeleton spec in this guide
Confidence bar	Progress	<code>value</code> set to the factor percentage	Used in Cell Popup and Advanced Analysis Panel factor rows. Radix's Progress primitive wires <code>role="progressbar" / aria-valuenow</code> declaratively. Track, fill, and height stay exactly as specced
Select dropdown	Select		Style per Form Controls spec in this guide
Multi-select dropdown	Popover + Checkbox	Custom composition, no native shadcn multi-select	Style per Multi-Select Dropdown spec in this guide. Trigger uses PopoverTrigger , panel uses PopoverContent
Checkbox	Checkbox		Style per Form Controls spec in this guide
Radio	RadioGroup + RadioGroupItem		Style per Form Controls spec in this guide
Route selection cards	RadioGroup + RadioGroupItem	Styled as full route cards, not the default dot control	Patrol Planner route alternatives are mutually exclusive, only one route is ever selected. Gives keyboard navigation between alternatives and announces the selected state.
Toggle switch	Switch		Style per Form Controls spec in this guide

DESIGN PATTERN	SHADCN COMPONENT	KEY PROPS / VARIANTS	NOTES
Slider	Slider	Track, range fill, and thumb are separate DOM elements inside the Radix primitive	Style with <code>.form-range-track</code> , <code>.form-range-fill</code> , <code>.form-range-thumb</code> tokens from this guide. Set <code>aria-valuetext</code> to a human-readable label (e.g., "14 May 2026")
Date picker	Calendar + Popover + PopoverTrigger	Native <code>input[type="date"]</code> is acceptable. Use Calendar + Popover when a custom inline calendar is required	Native inputs inherit <code>.form-input</code> styling. For the popover variant, the trigger button uses the outline button style (<code>.btn-outline</code>). The Calendar panel uses <code>—color-surface-raised</code> background with <code>—shadow-md</code>
Time picker	Native <code>input[type="time"]</code> or <code>input[type="datetime-local"]</code>		Inherits <code>.form-input</code> styling. No custom popover needed unless the project requires a specific time wheel UI
Tooltip	Tooltip + TooltipContent		Only for non-critical supplemental info.
Offline indicator	Popover	Click-triggered, not hover	Carries connectivity status, so it must not be a <code>Tooltip</code> per the rule above. Use <code>Popover</code> with <code>aria-expanded</code> on the trigger button. See Top Bar spec in this guide
Cell popup (map)	Popover	Custom positioning via MapLibre anchor	See Map Components section

DESIGN PATTERN	SHADCN COMPONENT	KEY PROPS / VARIANTS	NOTES
Dropdown menu	DropdownMenu		For action menus (context actions). Not for navigation
Text input	Input		Style per Form Controls spec in this guide
Textarea	Textarea		Style per Form Controls spec in this guide
Pagination	Pagination	PaginationContent, PaginationItem, PaginationLink, PaginationPrevious, PaginationNext, PaginationEllipsis	Style per Pagination spec in this guide
Filter chip (removable)	Badge	Custom composition, remove control is a nested icon Button	Style per Search and Filter Bar spec in this guide

Accessibility

The target is WCAG 2.2 Level AAA. Rangers use this app in the field, often with gloves in direct sunlight. Strong contrast is a requirement. Every interactive element has to

work with a keyboard, and colour is never the only way to communicate status.

PRINCIPLE

REQUIREMENTS IN THIS DESIGN SYSTEM

Perceivable

Body and primary text: 7:1 contrast minimum against its background (see table below). Secondary/muted text (—color-text-secondary) is held to large-text AA (3:1) rather than full AA (4.5:1) against —color-surface-bg, so per the colour decision table it is only used at —text-sm (14px) and up on surface-raised, and at 24px+ only on surface-bg, except on disabled/inactive controls, which are exempt from contrast requirements

Every image has descriptive alt text, decorative images use `alt=""`

Status shown by icon / label alongside colour, not colour alone

Operable

All functionality reachable by keyboard

Touch target: 44x44px

Focus order follows reading order, every interactive element shows a visible focus ring

Understandable

`lang="en"` on the html element, updated for any localised content

Form errors name the problem specifically ("Email is required", not "Invalid input")

Navigation is consistent across all pages

Labels are always visible, placeholder text does not replace labels

Robust

Correct ARIA roles and states on all custom components

Valid, semantic HTML

Contrast Ratios

TEXT COLOUR	BACKGROUND	RATIO	RESULT
#313131 body	#F2F2F2 surface-bg	11.6:1	AAA
#313131 body	#FFFFFF surface-raised	13.0:1	AAA
#003A6B heading	#F2F2F2 surface-bg	10.3:1	AAA
#003A6B heading	#FFFFFF surface-raised	11.5:1	AAA
#FFFFFF inverse	#003A6B surface-inverse	11.5:1	AAA
#FFFFFF inverse	#002354 surface-deep	15.3:1	AAA
#6897B8 decorative	#003A6B surface-inverse	3.6:1	Large text only
#8EADC4 light	#003A6B surface-inverse	4.9:1	AAA large text
#4F7392 muted	#F2F2F2 surface-bg	4.4:1	Large text only

Focus Ring Specification

Every interactive element gets a visible focus ring on keyboard navigation and selection.

LIGHT SURFACE	DARK SURFACE
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Focused button

Another element

Light bg: outline: 2px solid var(--brand-primary); outline-offset: 2px.

Dark bg: outline: 2px solid var(--color-text-inverse); outline-offset: 2px.

When writing a new component, use the rule that matches its surface. Define `:focus-visible` explicitly on each component.

Voice and Tone

The voice doesn't change between a patrol summary and a critical alert. What changes is the urgency and the length. We never sugarcoat, we never overstate.

WRITE

- 3 high severity incidents in Zone A-3.
- Report synced successfully.
- No tip-offs in the last 48 hours.
- GPS signal lost. Check device.
- Submit this report? It will be logged and synced to the server.

NEVER WRITE

- Oops! Something went wrong on our end.
- Your report was submitted!
- It looks like there are no tip-offs yet.
- We were unable to establish a GPS connection at this time.
- Are you sure you want to do this? This action cannot be undone!

CRITICAL ALERTS: ACTION FIRST

- Active poaching reported in Zone 4. Last updated 10:31 SAST.
- Signal strength critical.
- Sync failed. Tap to retry.

NEVER: PASSIVE, HEDGED, OR DRAMATIC

- There is some potential poaching activity detected.
- We're experiencing some difficulties syncing your data right now.
- Immediate action may be required!

Changelog

v2.0 : Complete refactor since Demo 1.

Savanna Sentinel